MATH 345 Skeleton Notes
Unit 1: Vectors, matrices, and linear maps
Section 1.1: Vectors

Vectors as lists of numbers

A real quantity is also called a $\boxed{\text{scalar}}$.

Definition: Vector An $\boxed{n\text{-vector}}$ is an ordered list of n scalars, often written in the form $\mathbf{v} = (v_1, v_2, \dots, v_n)$. The collection of all n-vectors is denoted $\mathbb{R}^n$.

We will also encounter vectors written as a row or column of numbers in square brackets, i.e.,

$$\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}.$$

It is customary to use bold letters for variables, such as $\mathbf{v}$ or $\mathbf{w}$, to represent vectors. We let $\mathbf{0}$ denote the vector $(0, \dots, 0)$ consisting only of zeros. A point in n-dimensional space can also be identified with an ordered list of n numbers using Cartesian coordinates. When referring specifically to a point P, we write $P(x_1, x_2, \dots, x_n)$ to indicate its coordinates.

Vectors as displacements in space

Vectors can be used to represent displacements between points in space. The vector $\mathbf{v} = (v_1, v_2, v_3)$ represents the displacement from $P(x_1, y_1, z_1)$ to

$Q(x_2, y_2, z_2)$, where $x_2 = x_1 + v_1$, $y_2 = y_1 + v_2$, and $z_2 = z_1 + v_3$. This vector is also written as $\overrightarrow{PQ}$. Because vectors often indicate displacements, they are drawn pictorially as arrows starting at a point, and ending where that point is displaced by the vector.

Activity: Vectors as displacements in space

What is the vector representing the displacement from the point $P(1, 2)$ to the point $Q(5, 7)$.

If $\mathbf{v} = (v_1, v_2)$ was the displacement, then we would have $(1 + v_1, 2 + v_2) = (5, 7)$, i.e., so that $1 + v_1 = 5$ and $2 + v_2 = 7$. Solving these equations gives $v_1 = 4$ and $v_2 = 5$, so that the displacement vector is $\mathbf{v} = (4, 5)$.

Definition: Length of a vector The length of a vector $\mathbf{v} = (v_1, \dots, v_n)$ is the quantity

$$\|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}.$$

Activity

Verify, using planar geometry, that the length of a 2-vector is the length of the line-segment of the arrow representing the vector.

Let $\mathbf{v} = (x, y)$ be an arbitrary 2-vector. Consider a triangle with vertices $P(0, 0)$, $Q(x, 0)$, and $R(x, y)$ (see textbook figure fig-1-1-triangle-pythagoras).

This triangle is right-angled (the two legs are horizontal and vertical), and the hypotenuse is the line-segment upon which the vector $\mathbf{v}$ is built. The length of the leg from $P(0, 0)$ to $Q(x, 0)$ is $|x|$, and the length of the leg from $Q(x, 0)$ to $R(x, y)$ is $|y|$. By Pythagoras' theorem, we conclude that the length of the hypotenuse is

$$\sqrt{|x|^2 + |y|^2} = \sqrt{x^2 + y^2} = \|\mathbf{v}\|.$$

In a calculus course, you may have mostly seen a vector as a displacement, but in this course it can also be a row of data or a list of model parameters, among other important applications. The same operations below will later compare documents, tokens, and feature vectors.

Adding and scaling vectors

Two displacements can be combined: just apply one displacement after the other is applied. This naturally leads to the notion of *adding* two vectors.

Definition: Adding vectors The sum of two n-vectors $\mathbf{v} = (v_1, \dots, v_n)$ and $\mathbf{w} = (w_1, \dots, w_n)$, denoted $\mathbf{v} + \mathbf{w}$, is the vector given by the tuple $(v_1 + w_1, \dots, v_n + w_n)$.

Remark The sum of two vectors can be illustrated pictorially as in textbook figure fig-1-1-adding-vectors by drawing the parallelogram with the vectors placed along two sides of the parallelogram.

One can also *scale* a vector by a given scalar quantity.

Definition: Scaling vectors If $\mathbf{v} = (v_1, \cdots, v_n)$ is an n-vector and $t \in \mathbb{R}$ is a scalar, we let $t\mathbf{v}$ denote the vector $(tv_1, \cdots, tv_n)$, i.e., multiplying each entry of the vector $\mathbf{v}$ by t.

Definition: Linear combinations and convex combinations of vectors If $\mathbf{v}_1, \dots, \mathbf{v}_k$ are vectors in $\mathbb{R}^n$ and $c_1, \dots, c_k$ are scalars, then

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$$

is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_k$. The scalars $c_1, \dots, c_k$ are the coefficients of the linear combination. If $c_i \geq 0$ for all i and $c_1 + \cdots + c_k = 1$, then the linear combination is a convex combination.

A convex combination can be interpreted as a weighted average.

Activity: A weighted average

Let $\mathbf{v}_1 = (10, 0)$, $\mathbf{v}_2 = (0, 10)$, $\mathbf{v}_3 = (10, 10)$, and $\alpha = (1/4, 1/4, 1/2)$. Compute $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3$.

We have

$$\frac{1}{4}(10, 0) + \frac{1}{4}(0, 10) + \frac{1}{2}(10, 10) = (7.5, 7.5).$$

The coefficients are nonnegative and add to 1, so this is a convex combination.

Dot products of vectors

Definition: Dot product The $\boxed{\text{dot product}}$ of two n-vectors $\mathbf{v} = (v_1, \dots, v_n)$ and $\mathbf{w} = (w_1, \dots, w_n)$ is the scalar quantity

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \cdots + v_n w_n.$$

Geometrically, the dot product of two vectors is a quantity which is related to the *angle* between the two vectors $\mathbf{v}$ and $\mathbf{w}$. If we draw the two vectors as arrows with the same starting point, then they form an angle on the plane containing both vectors, and the angle θ is given below.

Theorem: Angle between vectors For two nonzero vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, the $\boxed{\text{angle}}$ θ between the vectors is the quantity satisfying the equation

$$\cos\theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}.$$

The dot product can be thought of as a measure of the *similarity* between two vectors. Suppose θ is the angle between two vectors $\mathbf{v}$ and $\mathbf{w}$:

- If θ is close to zero, then the two vectors are close to pointing in the same direction. Since $\cos(0^\circ) = 1$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} \approx 1.$$

- If θ is close to 180°, then the two vectors are close to pointing in opposite directions. Since $\cos(180^\circ) = -1$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} \approx -1.$$

- If θ is close to 90°, the two vectors are close to pointing at right angles. Since $\cos(90^\circ) = 0$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} \approx 0.$$

  When $\mathbf{v} \cdot \mathbf{w} = 0$, so that $\theta = 90^\circ$, we say the two vectors are perpendicular, or $\boxed{\text{orthogonal}}$.

Thus the quantity

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|}$$

ranges between -1 and 1, and measures the degree to which the two vectors $\mathbf{v}$ and $\mathbf{w}$ point in the same, or opposite, directions. In some contexts, especially in certain areas of data science, this quantity is called the $\boxed{\text{cosine similarity}}$ between the vectors $\mathbf{v}$ and $\mathbf{w}$.

Warning Cosine similarity is undefined if either vector is the zero vector.

Activity

Calculate the angle between the two vectors $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (\sqrt{3} + 2, 2\sqrt{3} - 1)$.

We begin by calculating the lengths of the two vectors, i.e

$$\|\mathbf{v}\| = \sqrt{(1)^2 + (2)^2} = \sqrt{5}$$

and

$$\begin{aligned} \|\mathbf{w}\| &= \sqrt{(\sqrt{3} + 2)^2 + (2\sqrt{3} - 1)^2} \\ &= \sqrt{(3 + 4\sqrt{3} + 4) + (12 - 4\sqrt{3} + 1)} \\ &= \sqrt{20}. \end{aligned}$$

Next, we calculate the dot product

$$\begin{aligned} \mathbf{v} \cdot \mathbf{w} &= (1)(\sqrt{3} + 2) + (2)(2\sqrt{3} - 1) \\ &= \sqrt{3} + 2 + 4\sqrt{3} - 2 \\ &= 5\sqrt{3}. \end{aligned}$$

If θ is the angle between the two vectors, then

$$\begin{aligned}\cos(\theta) &= \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} \\ &= \frac{5\sqrt{3}}{\sqrt{5}\sqrt{20}} \\ &= \sqrt{3}/2.\end{aligned}$$

Thus the angle between the two vectors is

$$\theta = \cos^{-1}(\sqrt{3}/2) = 30^\circ.$$

Activity

Calculate the angle between the two vectors

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}.$$

We calculate that

$$\mathbf{v} \cdot \mathbf{w} = (2)(1) + (3)(0) + (0)(1) + (2)(-1) = 2 + 0 + 0 -$$

Thus if θ is the angle between the two vectors, then

$$\cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} = \frac{0}{\|\mathbf{v}\|\|\mathbf{w}\|} = 0.$$

Thus

$$\theta = \cos^{-1}(0) = 90^\circ.$$

Note Note that $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$ for any $\mathbf{v} \in \mathbb{R}^n$.

We will study the dot product in far more detail in Orthogonality and least squares and Abstract vector spaces and polynomial approximation.

Definition: Distance between vectors The Euclidean distance between two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\operatorname{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

It measures how far apart the endpoints are, while cosine similarity compares direction after normalization.

Applications of vectors

Vectors can represent quantities other than displacements. The entries of a vector can record measurements, counts, samples, or features. The meaning of a vector depends on what its coordinates represent.

- **Object:** Color
  Meaning of the entries: red, green, and blue intensities
- **Object:** Time series
  Meaning of the entries: measurements at successive times
- **Object:** Portfolio
  Meaning of the entries: amounts held in each asset
- **Object:** Image
  Meaning of the entries: pixel intensities, listed in a fixed order
- **Object:** Document
  Meaning of the entries: counts of selected words
- **Object:** Customer
  Meaning of the entries: purchases of selected products
- **Object:** Object with features
  Meaning of the entries: measured attributes such as size, price, weight, or rating

The same vector operations can have different interpretations. A sum can add purchases, add word counts, or add two time series. A scalar multiple can rescale an image, double a portfolio, or change units. A dot product can produce a score. A distance can compare two feature vectors. Cosine similarity compares direction after normalization.

Activity: Word-count vectors

Use the dictionary `linear`, `matrix`, `data`. A document vector records the number of times these words appear, in this order. The query

$$\mathbf{q} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

represents the phrase "linear data". Consider three document vectors

$$\mathbf{D}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \qquad \mathbf{D}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{D}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

Compute the cosine similarities and Euclidean distances between $\mathbf{q}$ and each $\mathbf{D}_i$. Rank the documents by cosine similarity.

First,

$$\begin{aligned} \operatorname{cosim}(\mathbf{q}, \mathbf{D}_1) &= \frac{\mathbf{q} \cdot \mathbf{D}_1}{\|\mathbf{q}\| \|\mathbf{D}_1\|} \\ &= \frac{4}{\sqrt{2}\sqrt{8}} \\ &= 1. \end{aligned}$$

Next,

$$\begin{aligned}\operatorname{cosim}(\mathbf{q}, \mathbf{D}_2) &= \frac{\mathbf{q} \cdot \mathbf{D}_2}{\|\mathbf{q}\|\|\mathbf{D}_2\|} \\ &= \frac{1}{\sqrt{2}\sqrt{2}} \\ &= \frac{1}{2}.\end{aligned}$$

Finally, $\operatorname{cosim}(\mathbf{q}, \mathbf{D}_3) = 0$. So the ranking by cosine similarity is

$$\mathbf{D}_1,\ \mathbf{D}_2,\ \mathbf{D}_3.$$

For the Euclidean distances,

$$\begin{aligned}
\operatorname{dist}(\mathbf{q}, \mathbf{D}_1) &= \|\mathbf{q} - \mathbf{D}_1\| \\
&= \left\| \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \right\| \\
&= \left\| \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix} \right\| = \sqrt{2}, \\
\operatorname{dist}(\mathbf{q}, \mathbf{D}_2) &= \|\mathbf{q} - \mathbf{D}_2\| \\
&= \left\| \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\| \\
&= \left\| \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\| = \sqrt{2}, \\
\operatorname{dist}(\mathbf{q}, \mathbf{D}_3) &= \|\mathbf{q} - \mathbf{D}_3\| \\
&= \left\| \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right\| \\
&= \left\| \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\| = \sqrt{6}.
\end{aligned}$$

The vector $\mathbf{D}_1$ points in exactly the same direction as $\mathbf{q}$ because $\mathbf{D}_1 = 2\mathbf{q}$. The vector $\mathbf{D}_2$ shares the word “linear” with the query but also contains “matrix”. The vector $\mathbf{D}_3$ contains only “matrix”, so it is orthogonal to the query.

From the previous activity, we see that vectors can have high cosine similarity while being relatively far in terms of Euclidean distance.

Activity: The same ranking in code

The following code repeats the cosine-similarity ranking from Word-count vectors.

```
import numpy as np

D1 = np.array([2, 0, 2], dtype=float)
D2 = np.array([1, 1, 0], dtype=float)
D3 = np.array([0, 2, 0], dtype=float)
q = np.array([1, 0, 1], dtype=float)

score1 = (D1 @ q) / (np.linalg.norm(D1) * np.linalg.norm(q))
score2 = (D2 @ q) / (np.linalg.norm(D2) * np.linalg.norm(q))
score3 = (D3 @ q) / (np.linalg.norm(D3) * np.linalg.norm(q))

score1, score2, score3
```

Output:

```
(1.0, 0.5, 0.0)
```

1. Which score corresponds to $\mathbf{D}_2$?
2. Where does the code compute $\mathbf{D}_1 \cdot \mathbf{q}$?
3. Which document is most similar to the query by cosine similarity?
4. Why does the output agree with the word-count vector activity?

The score for $\mathbf{D}_2$ is `score2`. The expression `D1 @ q` computes $\mathbf{D}_1 \cdot \mathbf{q}$. Since `score1` is largest, $\mathbf{D}_1$ is most similar to the query. The output agrees with the hand calculation: $1, \frac{1}{2}, 0$.

Example: Neural networks and transformers A neural network is a function built from layers. A layer takes numbers as input, combines them using weights, applies a rule, and sends output numbers to the next layer. In many neural networks, the input and output of a layer are vectors. The weights are adjusted from data during training.

A transformer is a neural-network architecture for sequences. In a language model, text is first broken into tokens. A token can be a word, part of a word, punctuation mark, or other text fragment. Each token is represented by a vector.

Activity: A tiny attention-style token-weighting calculation

Consider the token sequence

small, red, bird.

Suppose a transformer layer is updating the token "bird". Its $\boxed{\text{query}}$ vector represents what this token is looking for in the sequence. Each $\boxed{\text{key}}$ vector represents how another token can be matched by a query. Each $\boxed{\text{value}}$ vector is the vector that may be averaged into the updated representation.

For this activity, the score for token i is

$$\text{score}_i = \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_i.$$

Use

$$\mathbf{q}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{k}_{\text{small}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{k}_{\text{red}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{k}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the three scores

$$\mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{small}}, \qquad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{red}}, \qquad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{bird}}.$$

2. Normalize the positive scores by dividing each score by the sum of all three scores. Call the resulting weights

$$\alpha = \begin{bmatrix} \alpha_{\text{small}} \\ \alpha_{\text{red}} \\ \alpha_{\text{bird}} \end{bmatrix}.$$

3. Use the value vectors

$$\begin{bmatrix} 4 \end{bmatrix} \qquad \begin{bmatrix} 0 \end{bmatrix} \qquad \begin{bmatrix} 4 \end{bmatrix}$$

The scores are

$$\mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{small}} = 1, \qquad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{red}} = 1, \qquad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{bird}} =$$

The sum of the scores is

$$1 + 1 + 2 = 4.$$

Thus

$$\alpha = \begin{bmatrix} 1/4 \\ 1/4 \\ 1/2 \end{bmatrix}.$$

The weighted average is

$$\mathbf{h}_{\text{bird}} = \frac{1}{4}\begin{bmatrix} 4 \\ 0 \end{bmatrix} + \frac{1}{4}\begin{bmatrix} 0 \\ 4 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$

The token “bird” receives the largest weight because its key vector has the largest dot product with $\mathbf{q}_{\text{bird}}$. The updated vector is a weighted average of the value vectors.

Warning This toy calculation uses normalization by the sum of positive scores to keep the arithmetic simple. Real transformer attention usually uses a different weighting rule.